

Comparative Analysis for Prediction and Classification of Mental Health Issues and Challenges Using Hybrid Learning Techniques

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Abstract— Mental health condition including sadness, anxiety, and sleep deprivation owns up to the emotional stress in young children, teens, and also in adults. It affects how a person thinks, ponders, feels, or responds to a certain circumstance or situation. Only if one has both good physical health and mental health, an individual can work productively and reach their full potential. Mental health is very important at every stage of life, from childhood to adulthood. We gathered information from online datasets that were readily available. For better prediction, the data has been label-encoded. To obtain labels, the data is subjected to several machine-learning approaches. The model that will be developed to forecast a person's mental health will then be based on these categorized labels. Working-class individuals over the age of 18 are our target market. The model will be implemented into a website when it is created so that it may forecast the outcome based on the information provided by the user.

Keywords— Prediction, Classification, Hybrid Learning Algorithms, Machine Learning, Stress, Depression

I. INTRODUCTION

A. Overview

An individual's mental condition and overall nature can both be determined by that person's mental well-being. Unbalances in brain chemistry led to mental illness. To comprehend and recommend treatments for people with deviated mental conduct, it is essential to evaluate mental well-being. Most people are prone to stress, whereas other people experience depression for a variety of reasons. In 2011, a World Health Organization (WHO) administrative panel predicted that by 2030, depression will be the leading cause of disease burden globally. The delivery of better medication will necessitate a fundamental shift in which healthcare practitioners must now take into account the

mental fitness profile of the patient. The approach to machine learning is most frequently used in many different sorts of research, studies, and experiments, particularly in the medical field's ability to forecast illness. All data instances in supervised learning should reflect the names, characteristics, and values. Supervised learning, in more specific terms, is a classification method that makes use of structured training data. Unsupervised learning, on the other hand, does not require supervision to anticipate. Its primary objective is to handle data without supervision. The researchers' ability to use unsupervised learning techniques in the clinical setting is severely constrained.

The primary goals of this essay are to present a systematic literature analysis, a critical assessment, and a summary of Machine learning methods that are being utilized to anticipate, recognize, and diagnose mental health issues. This essay will also suggest potential directions for further study on the subject. Additionally, it would highlight the difficulties and restrictions associated with using machine learning techniques in this context. In addition, the future directions and gaps in this area of study will be highlighted. As a result, this research will advance the state of the art by doing a thorough literature assessment of the machine learning methods used to forecast mental health issues. Consequently, this work offers a critical overview and suggested future research areas.

B. Mental health

When you say health, both the mental health and physical health of the person fall into place. Both of them are equally important for a person to lead a healthy and joyful life. If the mental health of a person is not in the right state, it affects the person's

productivity, and thinking ability and even affects their physical health too. Mental health illness is one of the most common illnesses in the whole world. There is no specific reason for a person to have mental illness and so, one needs to be careful and watchful about their mental health. There are many types of mental illness, some of which are anxiety, bipolar disorder, depression, eating disorders, and stress. As serious as they sound, they are dangerous if a person has them. So, it is important to be mindful and careful about our mental health also as we are for our physical health.

C. Classification of mental illness

A mental disorder can be identified and classified by the disturbance in one's behavior or emotional regulation. There are a lot of different types of mental health illnesses. Some of them are,

- Anxiety disorders
- Depression
- Bipolar disorder
- Post-traumatic stress disorder
- Schizophrenia
- Eating disorders
- Neurodevelopmental disorder

D. Hybrid learning algorithms

Machine learning algorithms which are also known as black box algorithms work effectively but they do not let us get the full potential of AI. But on the other hand, hybrid learning methods come in handy. They are the combination of different small algorithms which complement each other and work together completely in great ways possible. We can either use two machine learning algorithms together or one machine learning and one deep learning algorithm together. So as a result, they eliminate uncertainty and give us more accurate results.

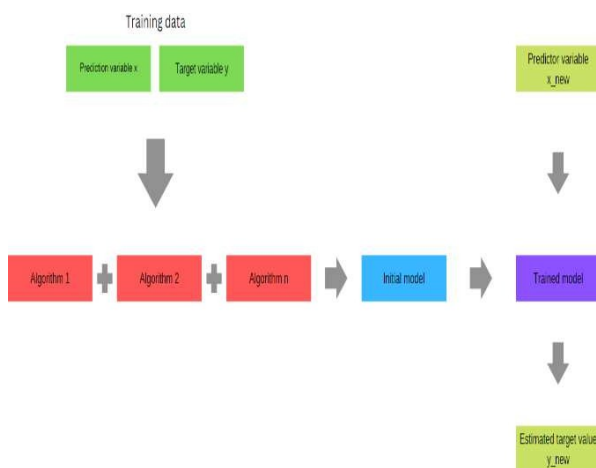


Fig.1. Working of the hybrid algorithm

There is a feature that helps computers to learn from their past and get even better and better going forward in time. This feature is known as machine learning and it is a subset of artificial intelligence. The foundation of machine learning is the creation of computer programs with the ability to retrieve data and acquire knowledge on their own. It reduces the workload of a human.

In the healthcare industry, where there is a vast amount of data, this is very useful since, when properly supplied to an intelligent system and trained suitably, the resulting prediction model will be unmatched, free from human errors, and shorten the time needed for diagnoses. There are a lot of other machine learning algorithms, some are used for classification, some for clustering, and some for prediction. Some of the prediction algorithms that can be of use to this project are,

1) Random forest classifier:

Random forest is one of the popular and well-known machine learning or ML algorithm. It falls under supervised machine learning algorithms. We can use the random forest in the places where we have to perform either classification or regression. Random forest algorithm has ensemble learning as the foundation because it is better if many classifiers or algorithms are integrated for better and enhanced performance algorithm is based on trees. It uses a combination of many possible trees on different data or subsets of data and finally sums up and finds the averages of the results. By this, we can achieve better accuracy and have the best results.

2) Support vector machine:

It is also well known as SVM. Alike random forest, SVM also can be used in both the cases of classification and regression. The SVM works by dividing the data into specific classes. It establishes the boundaries between the different groups or classes. This boundary is known as the hyperplane. This is the key factor of every SVM algorithm.

II. PROBLEM DEFINITION AND OBJECTIVE

A. Definition

People around the globe work hard to keep up with the racing world. As a result of this everyone is dealing with different health issues. One of the most crucial issues is mental health since it affects the person deeply. It is as important as physical health. So, it is important to predict any mental health issue such as stress in a person as soon as possible. That is the main purpose of this project.

B. Objective and workflow

Our objective here is to give predictions about the mental health of the people and classify them. We are also aiming to give a personal analysis of a person so that he/she can come to know about them in a more detailed manner. We are going to give predictions on as many mental health illnesses as possible in terms of stress or depression and also try and classify them as a low rate of stress, medium rate, and high rate.

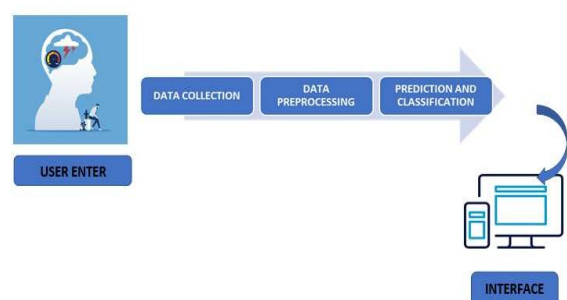


Fig.2. Workflow

The above figure shows our workflow. First, we are going to collect data, then preprocess it like removing any null values, encoding it, etc. and then finally apply algorithms to predict and classify mental health issues and provide an interface for ease of use and a user-friendly environment.

III. LITERATURE REVIEW

A diverse literature survey has been done for a better understanding of the methodologies involved in the prediction of mental health disease. A clear idea of how the algorithms work has been known after this. The literature review is a crucial part of any project development. The literature review done by us is as follows.

Sumathi et al [1] involve diagnosing basic mental health in children using several machine learning algorithms. The data are collected from the clinical psychologist and they have about 60 instances in text format. They have used eight algorithms that give good results for small datasets. In that dataset, only some of the features are taken into the consideration. Unnecessary features are left out. Necessary features are age, family history, academic performance, restlessness, learning difficulty, etc. These kinds of features are selected by the feature selection algorithm. The analysis of these algorithms is carried out by the WEKA tool. This tool helps us to understand the classification better. Also, they have used some measures like kappa statistics, accuracy, and ROC area to compare and analyze the accuracy level of classifiers and algorithms used.

Mutalib et al [2] described the subclass of linear algorithms known as SVMs (vector machines) that can be used to perform many tasks, such as classification, regression, density estimation, novelty detection, and others. SVM a forecasting model is used to develop a classification model. The SVM algorithm is created specifically to find a hyperplane that categorizes the data points in an N-dimensional space. When two classes of data points are separated, there could be a lot of potential hyperplanes. The Logistic Regression (LR) model offers the highest accuracy, specification, and precision values compared to other models, but a low sensitivity value. A Multilayer Perceptron (MLP) for ANN, on the other hand, has a lower accuracy of 68.89% and sensitivity of 60.00%, 75.00%, and greater specificity. 62.26%.2

Reddy et al [3] attempted in using ML models to find out the stress levels of the employees. They have collected the dataset from OSMI mental health survey. A lot of ML algorithms are used. Finally, they have concluded that KNN has the least accuracy and cannot be used for mental health detection.

Altrabseh et al. [4], the emotions of students are determined using the feedback given in the lectures taken by the professors. They have considered so many models for this prediction but using the CNB classifier, three of the models were finally used based on the three types of emotions. They are amused, excited, and bored. The total amount of feedback received was about 1522 tweets. There are four steps involved in using these algorithms. They are preprocessing, selecting the feature, applying the algorithm model, and finally obtaining the results. The algorithm uses naïve Bayes, support vector machine, and maximum entropy. Among many algorithms used, complement

naïve Bayes performs the best and gives the highest recall value for all.

In [5] Ansari L et.al. Different sentiment lexicons are used in the hybrid technique trials, and logistic regression (LR) is used for text categorization. DL techniques and lexicon-based models are combined to undertake the ensemble method experiments. Long short-term memory (LSTM) and Attentions are two recurrent neural networks (RNNs) that are used in this collection of experiments as part of the deep learning (DL) techniques. For all three datasets, the total result is more accurate than any individual LSTM or all-lexicon LR. Radiatuses yielded 0.77 F1 scores and a 75% accuracy rate. the sentient and all-lexicon LR models have an accuracy of 64% on the CLPsych dataset, but single- and bi-lexicon LR models have an accuracy of less than 60%. NRC SA, which achieves the maximum F1 with two of the three datasets, is the best feature among the single feature sets.

In [6] Single Y et. al The improved dataset is processed by KNN, Random Forest, and CNN-AdaBoost to create an ensemble learning model for stress prediction in the future. The developed model will then be used to forecast whether the final stress level will be Low, Medium, or High. Using stacking algorithms, we achieved the highest accuracy of about 88.86% and obtained the proper risk level prediction.

Tate et al. [7] developed a system for detecting the mental health problems of children in mid-adolescence. They have also checked if the ML models could give better results than Logistic regression. They concluded that using only machine learning algorithms is not sufficient for identifying mental health problems.

Maniyar A et al. [8] selects features based on which is having more effect on stress. According to that selected features are sex, family background, and the existence of medical advantage. Then they used machine learning algorithms to analyze the stress level. After that, they implemented some real-time applications for understanding the stress level by using some inputs directly given by the users. That application was developed using visual studio and SQL server. They used some supervised learning algorithms like KNN because it is a strong classifier, for a few and a large dataset, then performs well with a small and large number of parameters. And also classified into some training and testing datasets. Results are displayed to users by using the interface.

In [9] Eslamit mirjalili et.al. Various research studied the impact of various frequency bands on developing brain functional networks and found that Support Vector Machines (SVM) and random forests obtained 79.17%. By suggesting an element-wise layer for DNNs that included the data-driven structural priors, researchers were able to achieve 68.7% classification accuracy on 1, 013 participants, of which 539 were healthy controls and 474 had ASD. The ABIDE-I repository contains the datasets used in this investigation.

CO Alm et al. [10] produces the result from the text used by the users. They have used some supervised machine learning algorithms like SNoW learning architecture. Snow is an algorithm that can be used for multi-class classifiers which are specially meant good for working with a large number of attributes and features. This algorithm is mainly used in research studies as a research tool and also for several applications like

natural language and visual processing domains. Further, they also used a naïve base and BOW approach for classifying the emotional and non-emotional, positive and negative texts. The data are collected from the narrative stories of children's fairy tales. From that, they considered some features like the first sentence in the story, positive and negative word count, etc. They have achieved average accuracy rate, recall, and precision.

A Kene et al. [11] had developed a system for stress detection for some users. Data is collected in the form of a survey from 270 different people. The data is collected from a structured questionnaire. The attributes and classes in the dataset contain integer values. Users belong to the age group of 18-50. Data preprocessing have done to remove the null value data which is not necessary for stress detection. Then they implemented a support vector machine and random forest algorithm as classifier algorithms for classifying the feature as stress or no stress. They had also done some performance evaluations like precision, recall, and confusion matrix and got a good accuracy rate is 80.2%. This is achieved by using a support vector machine.

In [12] Shafiee et.al. The best method for classifying stress was created by CHAID Decision Tree (DT). Five levels make up the first goal variable: normal, mild, moderate, severe, and extremely severe. These levels are further divided into two stress types. 0 indicates that the kids' stress levels are normal or light, indicating that they do not have any mental health issues. Decision Trees and Support Vector Machines are the best models with the highest accuracy for stress and depression, respectively. The two models that provide reasonable results for anxiety have an accuracy range between 68% and 88%: linear regression and neural networks. In comparison to other models, the Logistic Regression (LR) model has the highest accuracy, specification, and precision values, but a low sensitivity value. During this time, a Multilayer Perceptron (MLP)

Kumar et al. [13] has used the dataset that has been collected in real-time from the working people through a questionnaire. Many ML algorithms have been applied to the data to find mental illness. They mainly focus on studying the despondency among the people. The used ML algorithms in this paper are Support vector machine, Random Forest, and Decision tree. Finally, they conclude by saying Random Forest is the best algorithm to use since it arrives at the highest accuracy comparatively.

Chung and Jason Teo [14] have also done a review of using ML algorithms to detect mental health. They have shown a clear review of the ML techniques for mental health detection by surveying many papers and also searching similar and trustable databases. They have researched a total of 30 papers. According to their research, deep learning algorithms have given the highest accuracy.

Priya et al. [15] have used 5 different machine learning models for detecting psychological health issues. Some of the issues are depression, stress, and anxiety. The data set that they have used is collected in real-time from the many employed and unemployed people. The best accuracy is found by the random forest algorithm.

Choudhury et al. [16] focuses mainly on Bangladeshi students and finds out why they are depressed. The data for this is found

through a survey that was designed specifically after consulting psychologists. They have used the random forest to find the features to predict mental illness and have also predicted depression using those features. They have also used DL algorithms, which as a result arrived at the best prediction with the best accuracy and f1 score.

Alharthi [17] tells that students who are prone to mental illness can be seen at the earliest stage possible by Artificial intelligence through machine learning. The data that she used is collected in real-time through an online survey. The best accuracy and the f1 score were both achieved by the deep learning algorithms. But the other algorithm called AUC or AdaBoost gave a better performance but a low accuracy rate and f1 score. So, she concluded that DL is a better method for stress detection or anxiety detection.

Nijhawan et al. [18] handles the data for cyberspace. They have used a large-scale dataset that consists of all the tweets and their objective are to predict the stress of a single person through the content shared by him on Twitter. They have used the deep learning model which is known as BERT and using BERT, for the sentient classification. They have also used some machine learning models. They conclude by saying that this can be very useful to know a person's mental health and well-being. They have given a result that the random forest classifier gives more accuracy than other ML algorithms.

Laijawala et al. [19] tells that it is important to identify mental illness problems at the beginning itself for it helps the people. They list several factors that indicate that the number is suffering from entering illness. The dataset for this has been collected from the internet. They have also done encoding for the data for better prediction.

Hossain E et al. [20] that persons with mental illness can be found using machine learning. They have also divided the approach into six parts. In the end, they find that the Random Forest classifier gives the best accuracy and can be used to find the mental illness of a person. They have also found that some quantity of people says 71% who had mental health issues can be diagnosed. So, they conclude by saying that this can be used on the clinical side and also, they plan to grow sentiment.

IV. SURVEY METHODOLOGY

A variety of measures will be taken in this research to create a mental health issues prediction model. The dataset will be analysed first, and the data will then be cleaned through pre-processing to provide the greatest performance during modelling. Finally, a mental health prediction model will be created using machine learning methods.

A. Data collection:

Some of the datasets were taken from the people in the form of online surveys, some datasets are collected from official hospitals or psychologists and some data are collected from online open platforms.

B. Data Cleaning:

In data cleaning, some parts will help to be performed after data exploration which will help in creating a more accurate machine learning model and eliminate bias after data cleaning

our data is composed and it will reduce the rows and columns. There is more advantage to data cleaning it will almost clear the unwanted data and unused data for a long time it will clean it and gives the correct input data. Finally, all missing values have been cleared to eliminate errors while building the models.

C. Model Building:

The Following Steps consist of so many stages that must be taken to start with splitting the data into training and testing data. Then, they applied machine learning techniques to build the model to compare the accuracy of this model. Some of them have also used a deep learning model and a machine learning model. The best accuracy was given by the hybrid and ensemble models and also by the Random Forest ML algorithm.

In some of the papers, they have also used an ML algorithm to know the best features for predicting the mental illness and then applied a DL algorithm for predicting the illness or applied both DL and ML algorithms on top of each other for best accuracy and prediction.

Table 1. Drawbacks

S.no	Description	Drawbacks
1	In this paper, 8 ML algorithms are used. Accuracy- 90% - LAD tree	Small dataset and data are not trained well
2	In this paper, ML methods are used to predict mental health issues – LR, Decision Tree, Random Forest, SVM, and neural network. Accuracy-65% - Decision tree	Low accuracy
3	In this paper, ML methods are used to predict the mental health issues – LR, KNN, Decision Tree, Random Forest, bagging, and boosting Accuracy-73% - Random Forest	A very small dataset is used.
4	This paper has used Naïve Bayes, Maximum Entropy, and Random Forest. Accuracy-67%	The predictions are not clear and the model is not trained in a good manner.
5	In this paper, ensemble methods are used to predict mental health issues – DL Lexicons and LR Accuracy-75%	Only depression is detected based on behavioral changes.
6	This paper uses CNN, KNN, Random Forest, CNN-AdaBoost Accuracy-88% -- Stacking algorithms	Provides results only on stress
7	In this paper, SVM, Neural networks, random forest, and XGBoost are used) Accuracy-75% - random forest	Small dataset
8	This paper uses the KNN algorithm for the detection Accuracy-87%	Only output that gives stressed employee or non-stressed employee
9	This paper uses ASD- Diagnet for the detection Accuracy-70.3%	No visualized result
10	In this paper, they have used the BoW method in NLP to predict emotions from text Accuracy -63%	The model had to be redefined one again
11	In this paper, they have used algorithms such as KNN, SVM, XGBoost, TVSM, Random Forest, and Naïve Bayes Accuracy- 80% -- SVM	Dataset is not defined well.
12	This paper has used a lot of ML algorithms such as SVM, Dt, Random Forest, Naïve Bayes, etc. Accuracy – 70 to 69% - SVM	Only depression is detected.

13	This paper has used SVM, Random Forest, and Decision tree to predict mental illness. Accuracy – 87.02%	Visual output is not that good.
14	This paper has also done a literature review by using the PRISMA methodology	--
15	In this paper, they have used SVM, DT, Random Forest, Naïve Bayes, and KNN to predict mental health issues Accuracy – 80% -- SVM	Had imbalanced partitioning
16	This paper uses deep learning, SVM, DT, Random Forest and so for detecting mental health issues. Accuracy- 63%	Non-uniformly distributed dataset.
17	This paper uses SVM, AdaBoost, Decision tree, Neural network, KNN, and neural networks for detecting mental health issues. Accuracy – 93%- neural network	Visualizations are not available
18	This paper has the BoW method from NLP and BERT model has also been used. Also, some ML algorithms such as random forest classifier, DT, and Logistic regression are used. Accuracy-97.7% - Random Forest classifier	Spams are not detected
19	This paper has used DT, Random Forest, Logistical regression, OneR, Naïve Bayes and NNge Accuracy – 82.2% - DT	The only stress is addressed
20	This paper has used Random Forest, XGBoost, and SVM Accuracy-92%	Small dataset

V. DISCUSSION AND OPEN ISSUES

In this project, we have implemented the prediction and classification of mental health issues. In the papers that we read, they have only predicted or classified them using ensemble or hybrid or machine learning, or deep learning.

We aim to predict, classify and provide good visualizations for the user to see and understand. We also aim to provide an interface for the user for easy navigation. Some of the issues in the papers are, they have not given clear output visualizations in the paper, and also the implementation is not elaborately explained [21-46].

VI. CONCLUSION

From this report, we can come to know about mental health and its impact on an individual's health. Its types and how important it is to be diagnosed and cured. What are hybrid algorithms and how one can apply them to find the mental illness in a person? Currently, mental health is a delicate and significant problem. It is essential for leading a balanced, healthy life. The state of one's mind affects their thoughts, actions, and emotions. It may have an impact on a person's performance and productivity. According to a WHO report, depression will be a leading contributor to mental disease worldwide, and individuals need to care more about their mental health to manage their personal and professional lives. Online outcome predictors can be used by those who are apprehensive to seek a diagnosis from a human. This algorithm will be tested after training, and a model will be created. It would then be made available on a website so that users could fill out the forms and receive the appropriate results.

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